

Deep Reinforcement Learning

Optimal Control

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Content

Content

- Recap of model-free RL
- Open-loop optimal control
 - Shooting versus collocation
 - Trajectory planning
 - Stochastic optimal control
 - The linear case
- Closed-loop control
 - Model Predictive Control (MPC)
 - Differentiable Predictive Control (DPC)

Where are we?

Chapter	Topic	Content
	Basics & tabular methods	
1-5	Bandits, MDPs, Dynamic Programming, Monte Carlo, TD Learning	RL basics in finite dimensions
	Deep-learning-based methods	
6-13	DQN, policy gradients, actor-critic, PPO/SAC, exploration	Deep RL from basics to modern algorithms
	Model-Based Control	
14	Optimal control	How to do control when we know the model
	Model-based reinforcement learning	
	Advanced Topics	

Lecture contents

Until now: model-free RL

What if we know the dynamics?

Planning & optimal control

The deterministic vs. the stochastic case

Open-loop versus closed-loop

Open loop optimal control

Open

Shooting versus collocation

Trajectory optimization

The linear case

Stochastic optimal control

Closed-loop control

Model predictive control (MPC)

Look at chapter 8 of (Sutton and Barto 1998)

Differentiable predictive control (DPC)

References

Sutton, R. S., and A. G. Barto. 1998. *Reinforcement Learning: An Introduction*. MIT Press.